

Experiment 3 :

```
import nltk

from nltk.tokenize import word_tokenize

from nltk.stem import PorterStemmer

# Download required resource (run once)

nltk.download('punkt')

ps = PorterStemmer()

text = input("Enter a sentence: ")

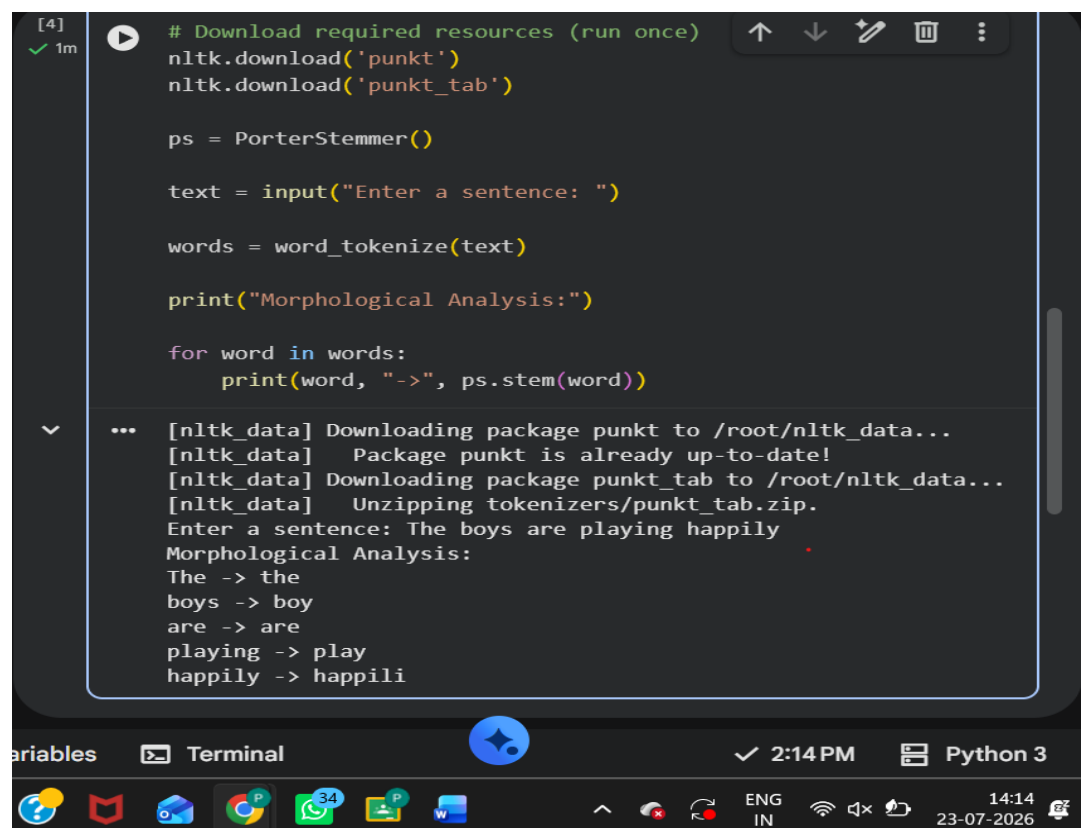
words = word_tokenize(text)

print("Morphological Analysis:")

for word in words:

    print(word, "->", ps.stem(word))
```

Output



The screenshot shows a Jupyter Notebook interface with a terminal window. The code in the terminal is the same as in the previous block. The output shows the download of NLTK resources, the creation of a PorterStemmer object, and the morphological analysis of the sentence "The boys are playing happily". The output is as follows:

```
[4] ✓ 1m # Download required resources (run once)
nltk.download('punkt')
nltk.download('punkt_tab')

ps = PorterStemmer()

text = input("Enter a sentence: ")

words = word_tokenize(text)

print("Morphological Analysis:")

for word in words:
    print(word, "->", ps.stem(word))

... [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
Enter a sentence: The boys are playing happily
Morphological Analysis:
The -> the
boys -> boy
are -> are
playing -> play
happily -> happili
```